

Practice Exam 2

Multiple Choice

1. A cranial nerve that does NOT innervate a muscle is the _____.
A. Facial nerve (VII) C. Olfactory nerve (I)
B. Accessory nerve (XI) D. Abducens nerve (VI)
E. Oculomotor nerve (III)
2. When do the posterior parietal cortex, premotor cortex, and supplementary motor cortex become most activated?
A. When planning a movement C. After a movement
B. During the movement itself D. None of the above
3. Within the visual system, typically many _____ converge onto several bipolar cells and _____ converge onto a single bipolar cell.
A. Rods;cones C. Cones;cones
B. Cones;rods D. Rods;rods
E. None of the above
4. In vision the parietal pathway is synonymous with the
A. "What" pathway C. Ventral pathway
B. Dorsal pathway D. None of the above
5. Axons from the nasal half of the retina go to the _____ hemisphere of the brain. Axons from the temporal half of the retina go to the _____ hemisphere of the brain.
A. Contralateral ... ipsilateral C. Ipsilateral ... ipsilateral
B. Contralateral ... contralateral D. Ipsilateral ... contralateral
6. Infections or bone growth that prevent the middle ear from transmitting sounds properly to the cochlea produce which type of deafness?
A. Nerve deafness C. Conduction deafness
B. Central deafness D. None of the above
7. Sound is converted to vibrations by which part of the middle ear _____.
A. Oval window C. Ossicles
B. Organ of corti D. Tympanic membrane
E. Pinna
8. The _____ is located in the precentral gyrus and the _____ is located in the postcentral gyrus.
A. Primary motor cortex; primary somatosensory cortex C. Premotor cortex; motor cortex
B. Primary somatosensory cortex; primary motor cortex D. Supplementary motor cortex; motor cortex

9. Small, unmyelinated, slow _____ fibers carry information from free nerve endings signaling _____ pain.
- A. A δ ... dull ...
 - B. C ... dull ...
 - C. A δ ... sharp ...
 - D. C ... sharp ...
10. Which sensory modality does not typically process information from the contralateral side?
- A. Vision
 - B. Touch
 - C. Pain
 - D. Olfaction
 - E. Gustation
11. The primary taste cortex is known as the:
- A. Hypothalamus
 - B. Flavor cortex
 - C. Orbitofrontal cortex
 - D. Insular cortex
12. The anterior 2/3 of the tongue sends info via which cranial nerve:
- A. Vagus nerve (X)
 - B. Glossopharyngeal (IX)
 - C. Facial nerve (VII)
 - D. All of the above
 - E. None of the above

Open response

*For each part, please answer using single words/phrases/
fill-in-the-blank/paragraphs/sketch as applicable.*

Motor System

1. (4 points) **Name** the two corticospinal tracts and how they differ based on where fibers decussate and what parts of the body they control.

2. (4 points) **What** two brain structures are part of the **extrapyramidal pathway** (that are not part of the pyramidal system)? **What motor function(s)** is each of these brain structures responsible for?

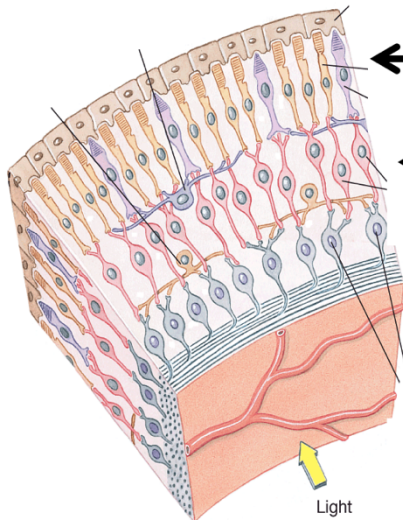
3. (10 points) You're a neurologist here at UNC hospital, and a patient referred to you has been diagnosed with Parkinson's disease. Please answer the following:
 - A) **Name** at least 2 symptoms she might be presenting with, that would have led to her diagnosis. (2pts)

 - B) **What midbrain region** and **neurotransmitter** system within the nervous system is degenerating in this condition and **what class of motor tracts** is being affected (pyramidal or extrapyramidal)? (4pts).

 - C) **Name** and **briefly describe** two treatment options that may be available to her for the management of her symptoms. (4pts)

Visual System

1. (9 points) **Cell types of the retina.** Label each cell layer on the diagram below.



A. _____

A B. _____

B C. _____

C There is an additional cell type found between layers A and B in the retina. What are these cells?

The axons from the cells in layer C form the _____ which exits out of the back of the eye and creates a _____ in the visual field.

2. (4 points) **Briefly describe** the functional and anatomic differences between rods and cones.

3. (6 points) **Describe** the key functions of the two major pathways in the visual cortex.

Auditory System

1. (5 points) **Name the correct order** for sound transduction in the ear: ossicles, auditory nerve, tympanic membrane, pinna, hair cells
2. (4 points) **Explain** the difference between middle-ear and inner-ear deafness.

Touch and pain

1. (3 points) How do hurt feelings resemble physical pain?
2. (4 points) **Briefly describe** the differences between pain and touch spinal pathways (Hint: ipsilateral vs. contralateral, where do fibers decussate?)
3. (4 points) **How** do opioids relieve pain and **why** does morphine relieve dull pain but not sharp pain?

Taste and smell

1. (6 points) **What** primary tastes do humans have? Name at least 3 and briefly describe the transduction mechanism of each taste receptor (Hint: ionotropic vs. metabotropic/G-protein-coupled).
2. (2 points) Given what is known about taste and smell, **why** does food taste different when a person has congested sinuses?
3. (3 points) **Briefly describe** the olfactory pathway to the orbitofrontal cortex.